

07770 428464
hajsteed@gmail.com

Harry Steed

github.com/hajsteed
linkedin.com/in/hajsteed

EDUCATION

Post-graduate research student (ongoing)

Apr 2025 - Oct 2028 (expected)

Cardiff University

Mathematical modelling and simulation of cell motion, with a focus on the relationship between local substrate topography and migration dynamics.

Master of Mathematics (Upper 2:1 with Honours)

Sep 2020 - Jun 2024

Cardiff University

Courses including: mathematical modelling, mathematical biology, differential equations, solid and fluid mechanics, and stochastic processes.

EXPERIENCE

Graduate tutor

Oct 2025 - Current

Cardiff University

Facilitating weekly problem classes for undergraduate students for Foundations of Mathematics I & II and Mechanics.

On-campus internship (Summer research project)

Jun 2023 - Aug 2023

Cardiff University

Researcher for an interdisciplinary project between Mathematics and Earth Science departments at Cardiff University.

- Developed a semi-automated software package for linear feature detection in biological images, replacing a previously manual process.
 - Employed signal processing techniques including wavelet and Fourier transforms for feature extraction.
 - Built an interactive user interface with buttons, sliders, and plots for non-technical researchers.
 - Designed and delivered a research presentation to a general audience at a departmental data jam.
-

EXTRACURRICULARS

Medical Imaging Hackathon

Jun 2025 - Jun 2025

- Developed a range of motion characteristics from single-cell and scratch assay microscopy videos; work led to ongoing collaboration with researchers at Reading University on a cell tracking pipeline.

Private Tutor

May 2018 - Jun 2023

- Managing individual students and planning personalised tutoring sessions for mathematical content ranging in difficulty from GCSE to degree level.

Rugby (Team Captain)

Sep 2014 - Aug 2018

- Led a team of peers during training sessions and games, helping to better organise team coordination and structure.
 - Responsible for split-second tactical decisions during a game.
-

SKILLS

- Programming: Matlab, Python, $\text{\LaTeX}$, Markdown, Git.
- Other: Stochastic simulation, spectral methods, image processing.